

Probability – I

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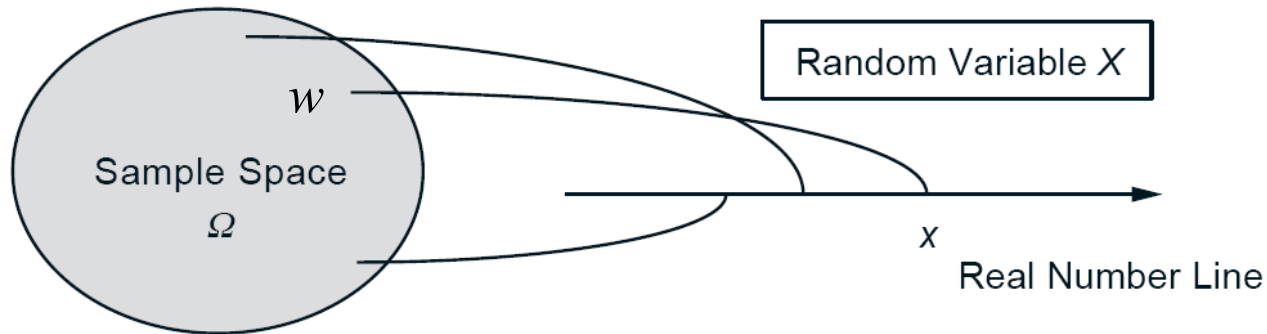
References:

- Dimitri P. Bertsekas and John N. Tsitsiklis, *Introduction to Probability*, Chapter 2
- Walpole R. E., Myers R. H., Myers S. L. and Ye K., *Probability & Statistics for Engineers & Scientists*, Ch. 3
- William Navidi, *Statistics for Engineers and Scientists*, Ch. 2.4
- Slides are credited from Prof. Berlin Chen, NTNU.

RANDOM VARIABLES: DISCRETE & CONTINUOUS

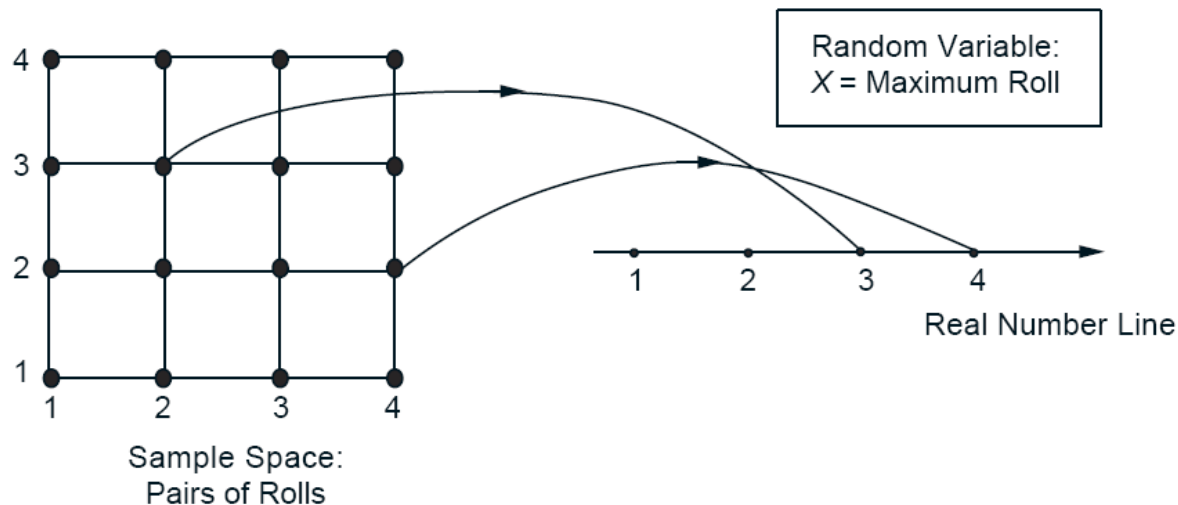
Random Variables

- Given an experiment and the corresponding set of possible outcomes (the sample space), a random variable associates a particular number with each outcome
 - This number is referred to as the (numerical) value of the random variable
 - We can say a random variable is a real-valued function of the experimental outcome



Random Variables: Example

- An experiment consists of two rolls of a 4-sided die, and the random variable is the **maximum** of the two rolls
 - If the outcome of the experiment is (4, 2), the value of this random variable is 4
 - If the outcome of the experiment is (3, 3), the value of this random variable is 3



- Can be one-to-one or many-to-one mapping

Main Concepts Related to Random Variables

- For a probabilistic model of an experiment
 - A random variable is **a real-valued function** of the outcome of the experiment

$$X : \omega \rightarrow x$$

- A **function of a random variable** defines another random variable

$$Y = g(x)$$

- We can associate with each random variable certain “averages” of interest such the **mean** and the **variance**
 - A random variable can be **conditioned** on an event or on another random variable
 - There is a notion of **independence** of a random variable from an event or from another random variable

Discrete/Continuous Random Variables

- A random variable is called **discrete** if its **range** (the set of values that it can take) is finite or at most countably infinite

finite : $\{1, 2, 3, 4\}$, countably infinite : $\{1, 2, \dots\}$

- A random variable is called **continuous** (not discrete) if its **range** (the set of values that it can take) is uncountably infinite
 - E.g., the experiment of choosing a point a from the interval $[-1, 1]$
 - A random variable that associates the numerical value a^2 to the outcome a is not discrete

Concepts Related to Discrete Random Variables

- For a probabilistic model of an experiment
 - A **discrete random variable** is a real-valued function of the outcome of the experiment that can take a finite or countably infinite number of values
 - A (discrete) random variable has an associated **probability mass function** (PMF), which gives the probability of each numerical value that the random variable can take
 - A **function of a random variable** defines another random variable, whose PMF can be obtained from the PMF of the original random variable

Probability Mass Functions

- A (discrete) random variable X is characterized through the probabilities of the values that it can take, which is captured by the **probability mass function (PMF)** of X , denoted $p_X(x)$

$$p_X(x) = \mathbf{P}(\{X = x\}) \text{ or } p_X(x) = \mathbf{P}(X = x)$$

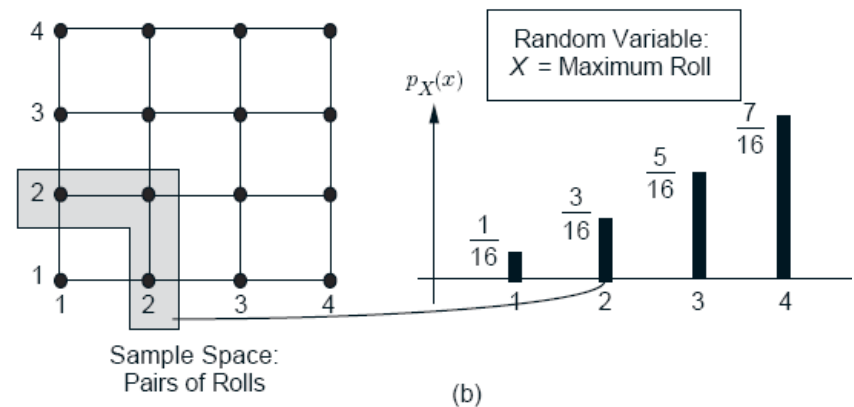
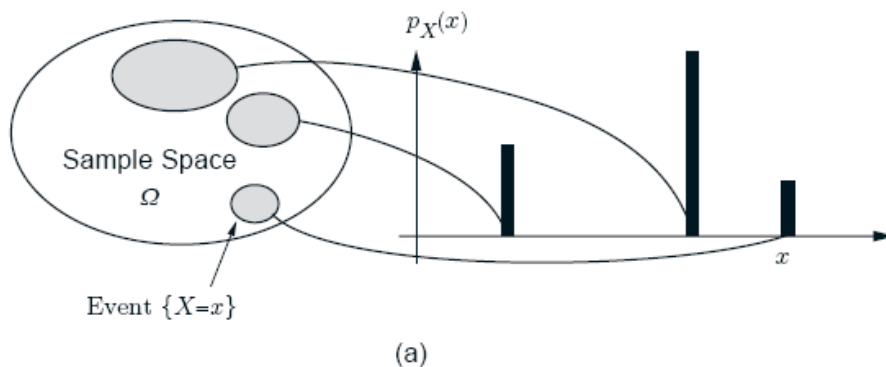
- The sum of probabilities of all outcomes that give rise to a value of X equal to x
- **Upper case** characters (e.g., X) denote random variables, while **lower case** ones (e.g., x) denote the numerical values of a random variable
- The probability mass function is sometimes called **probability distribution**

Cumulative Distribution Function

- The probability mass function specifies the probability that a random variable is equal to a given value
- A function called the **cumulative distribution function** (CDF) specifies the probability that a random variable is less than or equal to a given value
- The cumulative distribution function of the random variable X is the function $F(x) = \mathbf{P}(X \leq x)$

Calculation of the PMF

- For each possible value x of a random variable X :
 - Collect all the possible outcomes that give rise to the event $\{X = x\}$
 - Add their probabilities to obtain $p_X(x)$
- An example: the PMF $p_X(x)$ of the random variable $X =$ **maximum** roll in two independent rolls of a fair 4-sided die



More on Discrete Random Variables

- Let X be a **discrete** random variable. Then
 - The probability mass function (PMF) of X is the function
$$p_X(x) = \mathbf{P}(X = x)$$
 - The cumulative distribution function (CDF) of X is the function
$$F(x) = \mathbf{P}(X \leq x)$$
$$F(x) = \sum_{t \leq x} p_X(t) = \sum_{t \leq x} \mathbf{P}(X = t)$$
 - $\sum_x p_X(x) = \sum_x \mathbf{P}(X = x) = 1$, where the sum is over all the possible values of X

$\{X = x\}$'s are disjoint and form a partition of the sample space

Motivation for Expected Value (1/2)

- **An Illustrative Example:** Suppose that you spin the wheel k times, and that k_i is the number of times that the outcome (money) is m_i (there are n distinct outcomes, $m_1, m_2, \dots, m_n$)
- What is the amount of money that you “expect” to get “per spin”?
 - The total amount received is

$$m_1 k_1 + m_2 k_2 + \dots + m_n k_n$$

- The amount received per spin is

$$M = \frac{m_1 k_1 + m_2 k_2 + \dots + m_n k_n}{k}$$

Motivation for Expected Value (2/2)

- If the number of spins k is very large, and if we are willing to interpret probabilities as relative frequencies, it is reasonable to anticipate that m_i comes up a fraction of times that is roughly equal to p_i

$$p_i \approx \frac{k_i}{k}$$

- Therefore, the amount received per spin can be also represented as

$$\begin{aligned} M &= \frac{m_1 k_1 + m_2 k_2 + \cdots + m_n k_n}{k} \\ &= m_1 p_1 + m_2 p_2 + \cdots + m_n p_n \end{aligned}$$

Expectation

- The **expected value** (also called the **expectation** or the **mean**) of a random variable X , with PMF p_X , is defined by

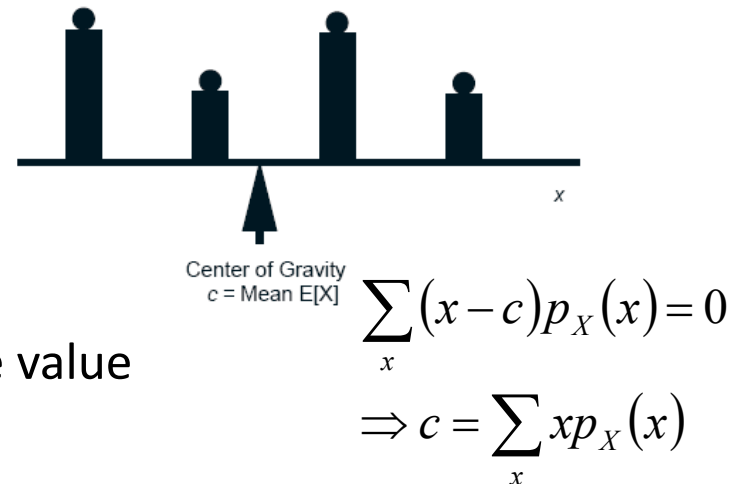
$$\mathbf{E}[X] = \sum_x x p_X(x)$$

- Can be interpreted as the **center of gravity** of the PMF (Or a weighted average, in proportion to probabilities, of the possible values of X)

- The expectation is well-defined if

$$\sum_x |x| p_X(x) < \infty$$

- That is, $\sum_x x p_X(x)$ converges to a finite value



Example

- **Example 2.2:** Consider two independent coin tosses, each with a $3/4$ probability of a head, and let X be the number of head obtained. This is a binomial random variable with parameters $n = 2$ and $p = 3/4$. Its PMF is

$$p_X(x) = \begin{cases} (1/4)^2, & \text{if } x = 0, \\ 2 \cdot (1/4) \cdot (3/4), & \text{if } x = 1, \\ (3/4)^2, & \text{if } x = 2, \end{cases}$$

$$\mathbf{E}(X) = \sum_x x p_X(x) = 0 \cdot \left(\frac{1}{4}\right)^2 + 1 \cdot \left(2 \cdot \frac{1}{4} \cdot \frac{3}{4}\right) + 2 \cdot \left(\frac{3}{4}\right)^2 = \frac{24}{16} = \frac{3}{2}$$

Moments

- The ***n*-th moment** of a random variable X is the expected value of a random variable X^n (or the random variable Y , $Y = g(X) = X^n$)

$$\mathbf{E}[X^n] = \sum_x x^n p_X(x)$$

- The 1st moment of a random variable X is just its mean (or expectation)

X^n is termed as X raised to the power of n (or the n th power) or the n th power of X

Variance

- The **variance** of a random variable X is the expected value of a random variable $(X - \mathbf{E}(X))^2$

$$\begin{aligned}\text{var}(X) &= \mathbf{E}[(X - \mathbf{E}[X])^2] \\ &= \sum_x (x - \mathbf{E}[X])^2 p_X(x)\end{aligned}$$

- The variance is always nonnegative (why?)
- The variance provides a measure of dispersion of X around its mean
- The standard derivation is another measure of dispersion, which is defined as (a square root of variance)

$$\sigma_X = \sqrt{\text{var}(X)}$$

- Easier to interpret, because it has the same units as X

Variance in Terms of Moments Expression

- We can also express variance of a random variable X as

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$$

$$\begin{aligned}\text{var}(X) &= \sum_x (x - \mathbf{E}[X])^2 p_X(x) \\ &= \sum_x (x^2 - 2x\mathbf{E}[X] + (\mathbf{E}[X])^2) p_X(x) \\ &= \left[\sum_x x^2 p_X(x) \right] - 2\mathbf{E}[X] \left[\sum_x x p_X(x) \right] + (\mathbf{E}[X])^2 \left[\sum_x p_X(x) \right] \\ &= \mathbf{E}[X^2] - 2(\mathbf{E}[X])^2 + (\mathbf{E}[X])^2 \\ &= \mathbf{E}[X^2] - (\mathbf{E}[X])^2\end{aligned}$$

Mean and Variance for Discrete Random Variables

- The **mean** (or expected value) of X is given by

$$\mu_X = \sum_x x p_X(x) \quad , \text{ also denoted as } \mathbf{E}[X]$$

where the sum is over all possible values of X

- The **variance** of X is given by

$$\begin{aligned} \sigma_X^2 &= \sum_x (x - \mu_X)^2 p_X(x) \quad , \text{ also denoted as } \mathbf{E}[(X - \mu_X)^2] \\ &= \sum_x x^2 p_X(x) - \mu_X^2 \quad , \text{ also denoted as } \mathbf{E}[X^2] - (\mathbf{E}[X])^2 \end{aligned}$$

- The **standard deviation** is the square root of the variance

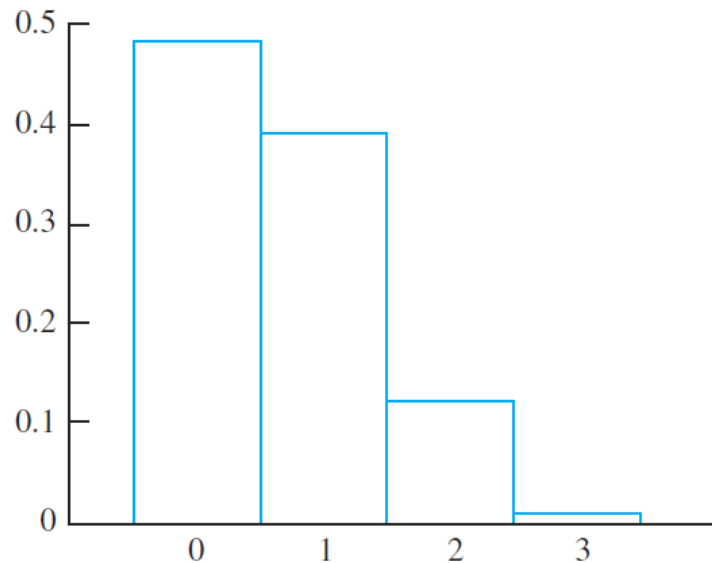
The Probability Histogram

- When the possible values of a discrete random variable are evenly spaced, the probability mass function can be represented by a histogram, with rectangles centered at the possible values of the random variable
- The area of the rectangle centered at a value x is equal to $\mathbf{P}(X = x)$
- Such a histogram is called a **probability histogram**, because the areas represent probabilities

Example

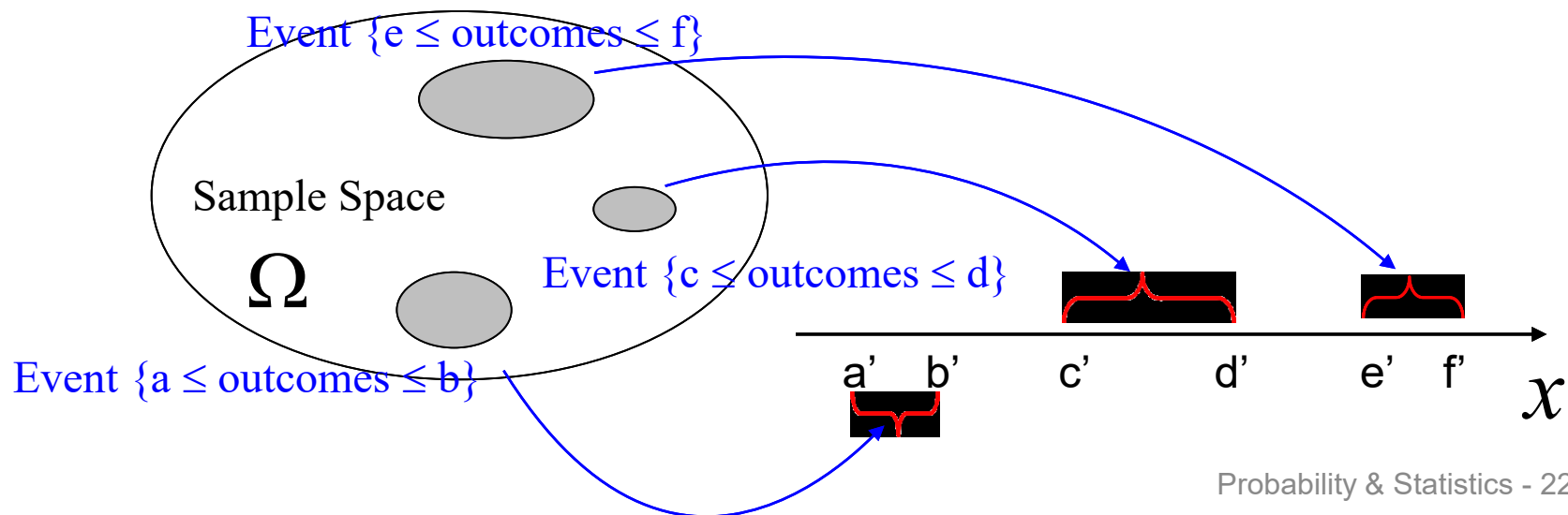
- The following is a probability histogram for the example with number of flaws in a randomly chosen piece of wire
 - $P(X = 0) = 0.48$, $P(X = 1) = 0.39$, $P(X = 2) = 0.12$, $P(X = 3) = 0.01$

- Figure 2.10



Continuous Random Variables (1/2)

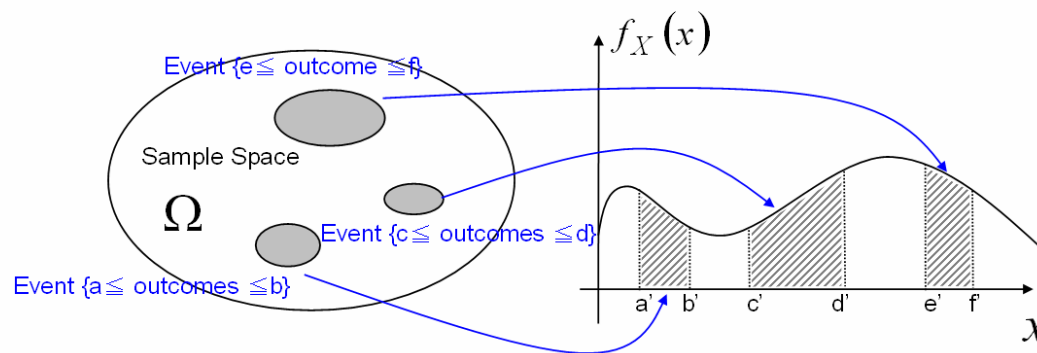
- Random variables with a continuous range of possible values are quite common
 - The velocity of a vehicle
 - The temperature of a day
 - The blood pressure of a person
 - etc.



Continuous Random Variables (2/2)

- A random variable is **continuous** if its probabilities are given by areas under a curve
- The curve is called a **probability density function (PDF)** for the random variable. Sometimes the PDF is called the **probability distribution**
- Let X be a continuous random variable with probability density function $f(x)$. Then

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$



Computing Probabilities

- Let X be a continuous random variable with probability density function $f(x)$. Let a and b be any two numbers, with $a < b$. Then

$$\begin{aligned}\mathbf{P}(a \leq X \leq b) &= \mathbf{P}(a < X \leq b) = \mathbf{P}(a \leq X < b) = \mathbf{P}(a < X < b) \\ &= \int_a^b f(x)dx\end{aligned}$$

- In addition,

$$\mathbf{P}(X \leq a) = \mathbf{P}(X < a) = \int_{-\infty}^a f(x)dx$$

$$\mathbf{P}(X \geq a) = \mathbf{P}(X > a) = \int_a^{\infty} f(x)dx$$

Cumulative Distribution Functions

- The **cumulative distribution function** (CDF) of a random variable X is denoted by $F_X(x)$ and provides the probability $\mathbf{P}(X \leq x)$

$$F_X(x) = \mathbf{P}(X \leq x) = \begin{cases} \sum_{k \leq x} p_X(k), & \text{if } X \text{ is discrete} \\ \int_{-\infty}^x f_X(t) dt, & \text{if } X \text{ is continuous} \end{cases}$$

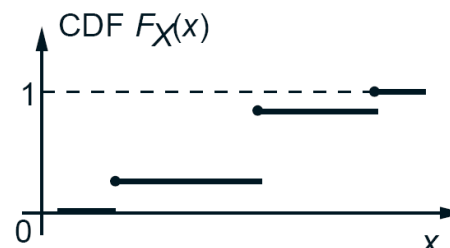
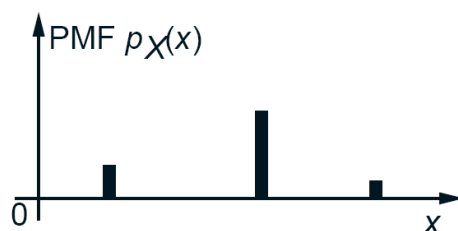
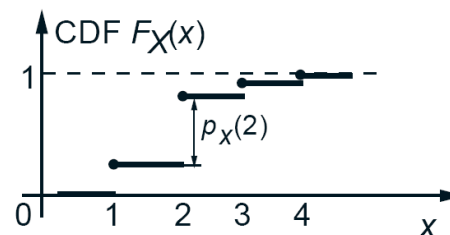
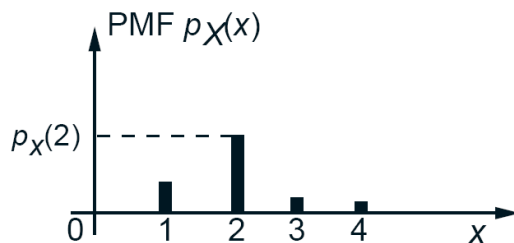
- The CDF $F_X(x)$ accumulates probability up to x
- The CDF $F_X(x)$ provides **a unified way** to describe all kinds of random variables mathematically

Properties of a CDF (1/3)

- The CDF $F_X(x)$ is **monotonically non-decreasing**

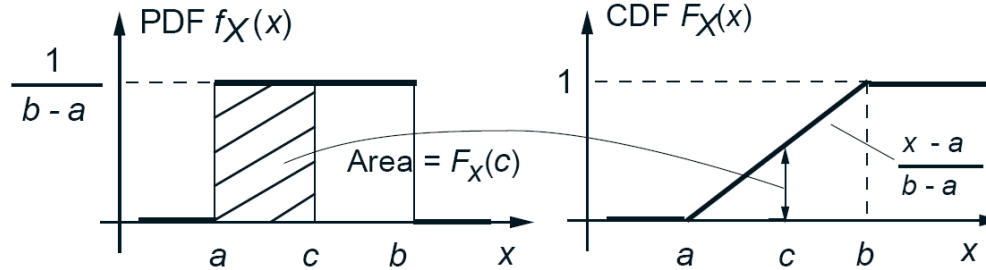
$$\text{if } x_i \leq x_j, \text{ then } F_X(x_i) \leq F_X(x_j)$$

- The CDF $F_X(x)$ tends to 0 as $x \rightarrow -\infty$, and to 1 as $x \rightarrow \infty$
- If X is **discrete**, then $F_X(x)$ is a **piecewise constant** function of x



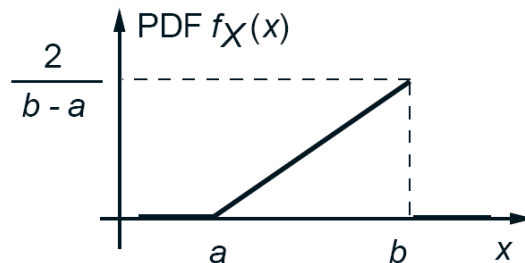
Properties of a CDF (2/3)

- If X is continuous, then $F_X(x)$ is a continuous function of x



$$f_X(x) = \frac{1}{b-a}, \text{ for } a \leq x \leq b$$

$$F_X(X \leq x) = \int_a^x f_X(t) dt = \int_a^x \frac{1}{b-a} dt$$

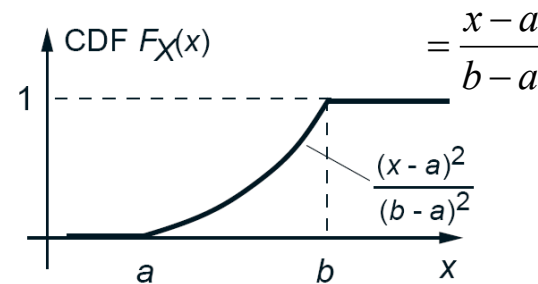


$$f_X(x) = c(x-a), \text{ for } a \leq x \leq b$$

$$\Rightarrow \int_a^b c(x-a) dx = \frac{c}{2} (x-a)^2 \Big|_a^b = 1$$

$$\Rightarrow c = \frac{2}{(b-a)^2}$$

$$\Rightarrow f_X(b) = \frac{2(b-a)}{(b-a)^2} = \frac{2}{b-a}$$



$$F_X(X \leq x) = \int_a^x f_X(t) dt = \int_a^x \frac{2(t-a)}{(b-a)^2} dt$$

$$= \frac{(x-a)^2}{(b-a)^2}$$

Properties of a CDF (3/3)

- If X is **discrete** and takes integer values, the PMF and the CDF can be obtained from each other by summing or differencing

$$F_X(k) = \mathbf{P}(X \leq k) = \sum_{i=-\infty}^k p_X(i),$$

$$p_X(k) = \mathbf{P}(X \leq k) - \mathbf{P}(X \leq k-1) = F_X(k) - F_X(k-1)$$

- If X is **continuous**, the PDF and the CDF can be obtained from each other by integration or differentiation

$$F_X(x) = \mathbf{P}(X \leq x) = \int_{-\infty}^x f_X(t) dt,$$

$$\mathbf{P}(a < X < b) = \int_a^b f_X(x) dx$$

$$f_X(x) = \frac{dF_X(x)}{dx}$$

is equivalent to ?

$$= F_X(b) - F_X(a)$$

- The second equality is valid for those x for which the CDF has a derivative (or at which PDF is continuous)

Example

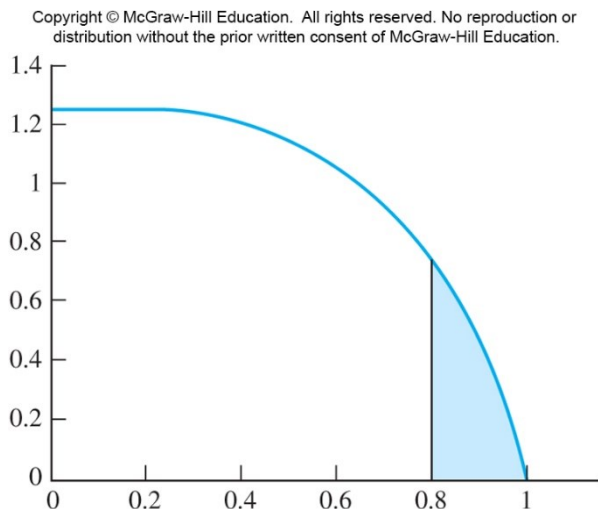
- Example 2.41: A hole is drilled in a sheet-metal component, and then a shaft is inserted through the hole. The shaft clearance is equal to the difference between the radius of the hole and the radius of the shaft. Let the random variable X denote the clearance, in millimeters. The probability density function of X is

$$f(x) = \begin{cases} 1.25(1 - x^4) & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

1. Components with clearances larger than 0.8 mm must be scrapped. What proportion of components are scrapped?
2. Find the cumulative distribution function $F(x)$.

Answer

1. Figure 2.13: Graph of the PDF of X



$$\begin{aligned}\mathbf{P}(X > 0.8) &= \int_{0.8}^{\infty} f_X(x) dx \\ &= \int_{0.8}^{\infty} 1.25(1 - x^4) dx \\ &= 1.25 \left(x - \frac{x^5}{5} \right) \bigg|_{0.8}^1 \\ &= 0.0819\end{aligned}$$

$$2. \quad \text{If } x \leq 0 \quad F(x) = \int_{-\infty}^x f(t) dt$$

$$= \int_{-\infty}^x 0 dt$$

$$= 0$$

$$\text{If } 0 < x < 1 \quad F(x) = \int_{-\infty}^x f(t) dt$$

$$= \int_{-\infty}^0 f(t) dt + \int_0^x f(t) dt$$

$$= \int_{-\infty}^0 0 dt + \int_0^x 1.25(1 - t^4) dt$$

$$= 0 + 1.25 \left(t - \frac{t^5}{5} \right) \Big|_0^x$$

$$= 1.25 \left(x - \frac{x^5}{5} \right)$$

$$F(x) = \begin{cases} 0, & x \leq 0 \\ 1.25 \left(x - \frac{x^5}{5} \right), & 0 < x < 1 \\ 1, & x \geq 1 \end{cases}$$

$$\text{If } x \geq 1$$

$$F(x) = \int_{-\infty}^x f(t) dt$$

$$= \int_{-\infty}^0 f(t) dt + \int_0^1 f(t) dt + \int_1^x f(t) dt$$

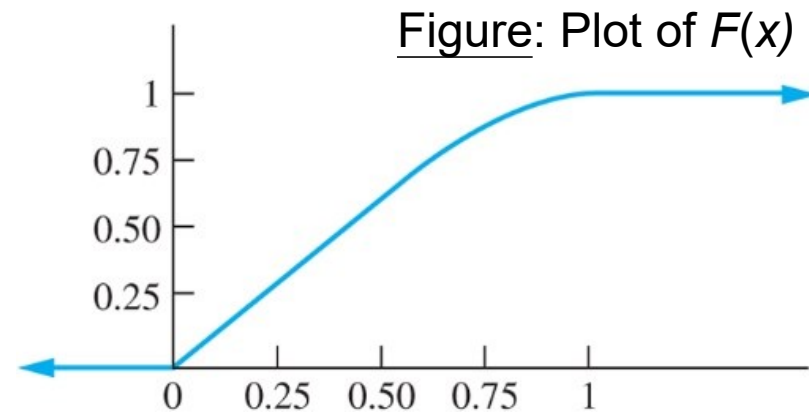
$$= \int_{-\infty}^0 0 dt + \int_0^1 1.25(1 - t^4) dt + \int_1^x 0 dt$$

$$= 0 + 1.25 \left(t - \frac{t^5}{5} \right) \Big|_0^1 + 0$$

$$= 0 + 1 + 0$$

$$= 1$$

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Another Example on Continuous RV

- Suppose that the error in the reaction temperature, in °C (實驗反應溫度), for a controlled laboratory experiment is a continuous random variable X having the probability density function

$$f_X(x) = \begin{cases} \frac{x^2}{3}, & \text{if } -1 < x < 2, \\ 0, & \text{otherwise,} \end{cases}$$

- Problems:
 - a. Find $\mathbf{P}(0 < X \leq 1)$.
 - b. Find the $F(x)$, and use it to evaluate $\mathbf{P}(0 < X \leq 1)$
 - Solution:

- a. $\mathbf{P}(0 < x \leq 1) = \int_0^1 f_X(x) dx = \int_0^1 \frac{x^2}{3} dx = \frac{x^3}{9} \Big|_0^1 = \frac{1}{9}$

- b. $F_X(x) = \mathbf{P}(X \leq x) = \int_{-\infty}^x f_X(t) dt = \frac{t^3}{9} \Big|_{-1}^x = \frac{x^3 + 1}{9}$

$$\mathbf{P}(0 < x \leq 1) = F(1) - F(0) = \frac{2}{9} - \frac{1}{9} = \frac{1}{9}$$

$$F_X(x) = \begin{cases} 0, & x \leq -1 \\ \frac{x^3 + 1}{9}, & -1 < x < 2, \\ 1, & x \geq 2. \end{cases}$$

Exercise #13

- Exercise #13: Resistor(電阻器) labeled $100\ \Omega$ have true resistances that are between $80\ \Omega$ and $120\ \Omega$. Let X be the mass of a randomly chosen resistor. The probability density function of X is given by

$$f(x) = \begin{cases} \frac{x-80}{800} & 80 < x < 120 \\ 0 & \text{otherwise} \end{cases}$$

1. What proportion of resistors have resistances less than $90\ \Omega$?

Ans: 1/16

2. Find the cumulative distribution function $F(x)$ of the resistances.

Ans:

$$F(x) = \begin{cases} 0, & x \leq 80 \\ \frac{x^2}{1600} - \frac{x}{10} + 4, & 80 < x < 120 \\ 1, & x \geq 120 \end{cases}$$

Exercise #18

- Exercise #18: The lifetime, in years, of a certain type of pump is a random variable with probability density function

$$f(x) = \begin{cases} \frac{64}{(x+2)^5} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

1. What is the probability that a pump lasts more than two years?

Ans: 1/16

2. Find the cumulative distribution function $F(x)$ of the lifetime.

Ans:

$$F(x) = \begin{cases} 1 - \frac{16}{(x+2)^4}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

More on Continuous Random Variables

- Let X be a continuous random variable with probability density function $f(x)$. The **cumulative distribution function (CDF)** of X is the function

$$F(x) = \mathbf{P}(X \leq x) = \int_{-\infty}^x f(t)dt,$$

- The **mean** (or expected value) of X is given by

$$\mu_X = \int_{-\infty}^{\infty} xf(x)dx \quad , \text{ also denoted as } \mathbf{E}[X]$$

- The **variance** of X is given by

$$\begin{aligned} \sigma_X^2 &= \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x)dx \quad , \text{ also denoted as } \mathbf{E}[(X - \mu_X)^2] \\ &= \int_{-\infty}^{\infty} x^2 f(x)dx - \mu_X^2 \quad , \text{ also denoted as } \mathbf{E}[X^2] - (\mathbf{E}[X])^2 \end{aligned}$$

Example

$$f(x) = \begin{cases} 1.25(1-x^4) & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

- Example 2.44: Refer to Example 2.41. Find the **mean** clearance and the **variance** of the clearance.

1. The mean is given by

$$\begin{aligned} \mu_X &= \int_{-\infty}^{\infty} xf(x)dx \\ &= \int_0^1 x[1.25(1-x^4)]dx \\ &= 1.25 \left(\frac{x^2}{2} - \frac{x^6}{6} \right) \Big|_0^1 \\ &= 0.4167 \end{aligned}$$

2. Having computed $\mu_X = 0.4167$, we can now compute σ_X^2 by using the equation:

$$\sigma_X^2 = \int_{-\infty}^{\infty} x^2 f(x)dx - \mu_X^2$$

$$\begin{aligned} \sigma_X^2 &= \int_{-\infty}^{\infty} x^2 f(x)dx - \mu_X^2 \\ &= \int_0^1 x^2 [1.25(1-x^4)]dx - (0.4167)^2 \\ &= 1.25 \left(\frac{x^3}{3} - \frac{x^7}{7} \right) \Big|_0^1 - (0.4167)^2 \\ &= 0.0645 \end{aligned}$$

Exercise #13

$$f(x) = \begin{cases} \frac{x-80}{800} & 80 < x < 120 \\ 0 & \text{otherwise} \end{cases}$$

- Exercise #13: Find the **mean** resistance and the **standard deviation** of the resistances.

1. The mean is given by

$$\begin{aligned} \mu_X &= \int_{-\infty}^{\infty} xf(x)dx \\ &= \int_{80}^{120} x \left(\frac{x-80}{800} \right) dx \\ &= \left(\frac{x^3 - 120x^2}{2400} \right) \Big|_{80}^{120} = \frac{320}{3} = 106.67 \end{aligned}$$

2. Having computed $\mu_X = 106.67$, we can now compute $\sigma_X = \sqrt{\sigma_X^2}$:

$$\begin{aligned} \sigma_X^2 &= \int_{-\infty}^{\infty} x^2 f(x) dx - \mu_X^2 \\ &= \int_{80}^{120} x^2 \left(\frac{x-80}{800} \right) dx - \left(\frac{320}{3} \right)^2 \\ &= \int_{80}^{120} \left(\frac{x^3}{800} - \frac{x^2}{10} \right) dx = \left(\frac{x^4}{3200} - \frac{x^3}{30} \right) \Big|_{80}^{120} - \left(\frac{320}{3} \right)^2 = \frac{800}{9} \end{aligned}$$

$$\sigma_X = \sqrt{\sigma_X^2} = \sqrt{\frac{800}{9}} = 9.428$$

Median and Percentiles

- Let X be a continuous random variable with probability mass function $f(x)$ and cumulative distribution function $F(x)$
 - The **median** of X is the point x_m that solves the equation

$$F(x_m) = \mathbf{P}(X \leq x_m) = \int_{-\infty}^{x_m} f(x)dx = 0.5$$

- If p is any number between 0 and 100, the **p th percentile** is the point x_p that solves the equation

$$F(x_p) = \mathbf{P}(X \leq x_p) = \int_{-\infty}^{x_p} f(x)dx = p / 100$$

- **The median is the 50th percentile**

Exercise #18

$$f(x) = \begin{cases} \frac{64}{(x+2)^5} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

- Exercise #18:

1. What is the probability that a pump last between two and four years?

2. Find the mean lifetime.

$$\begin{aligned} \mu_X &= \int_0^{\infty} x \frac{64}{(x+2)^5} dx = \int_2^{\infty} (u-2) \frac{64}{u^5} du = 64 \int_2^{\infty} (u^{-4} - 2u^{-5}) du \\ &= 64 \left(-\frac{1}{3} u^{-3} + \frac{1}{2} u^{-4} \right) \Big|_2^{\infty} = 2/3 \end{aligned}$$

3. Find the variance of the lifetimes.

$$\begin{aligned} \sigma_X^2 &= \int_0^{\infty} x^2 \frac{64}{(x+2)^5} dx - \mu_X^2 \\ &= \int_2^{\infty} (u-2)^2 \frac{64}{u^5} du - (2/3)^2 \\ &= 64 \int_2^{\infty} (u^{-3} - 4u^{-4} + 4u^{-5}) du - 4/9 \\ &= 64 \left(-\frac{1}{2} u^{-2} + \frac{4}{3} u^{-3} - u^{-4} \right) \Big|_2^{\infty} - 4/9 = \frac{8}{9} \end{aligned}$$

Exercise #18

$$f(x) = \begin{cases} \frac{64}{(x+2)^5} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

- Exercise #18:

1. What is the probability that a pump last between two and four years?

$$\mathbf{P}(2 < X < 4) = \int_2^4 \frac{64}{(x+2)^5} = -\frac{16}{(x+2)^4} \Big|_2^4 = \frac{65}{1296}$$

2. Find the mean lifetime. (next)
3. Find the variance of the lifetimes. (next)

$$F(x) = \begin{cases} 1 - \frac{16}{(x+2)^4}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

4. Find the median lifetime. The median x_m solves $F(x_m) = 0.5$.

Therefore $1 - \frac{16}{(x_m + 2)^4} = 0.5$, so $x_m = 0.3784$

5. Find the 60th percentile of the lifetimes. The 60th percentile x_{60} solves $F(x_{60}) = 0.6$. Therefore $1 - \frac{16}{(x_{60} + 2)^4} = 0.6$, so $x_{60} = 0.5149$